

MASOUMEH ALINIA

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EDUCATION

ALZAHRA UNIVERSITY, Tehran, Iran

February 2022-September 2024

Master of Science (M.Sc.), Software Engineering | GPA: 18.57/20 (ranked 2nd in the whole program) | Thesis Vqr le " Collaborative filtering recommender systems with deep learning approach | Advisor: Dr. Hasheminejad

SHAHID BEHESHTI UNIVERSITY, Tehran, Iran

October 2017-October 2020

Master of Science (M.Sc.), Electronic Engineering, Specialization in Digital Electronic | GPA: 16.74/20 | Vj gukt'Vqr le " Vj tgg-Valued Memory Based on Nanoscale Uf lptqple"Vgej pqiqi ""Cf xkqt "Dr. Timarchi

TECHNICAL AND VOCATIONAL UNIVERSITY, Bojnurd and Mashhad, Iran

October 2012-July 2016

Bachelor of Science (B.Sc.), Electronic Engineering | GPA: 18.87/20

PROFESSIONAL EXPERIENCE

ARMAN (FinTech Arm of Export Development Bank of Iran), Tehran, Iran (Full-Time)

October 2025-Present

Arman Development and Technologies (Arman) is a subsidiary of the Export Development Bank of Iran (**EDBI**), streamlining IT services for the bank and its other subsidiaries. Arman's primary objective is to develop, implement, and support software and hardware infrastructures to ensure efficient and agile delivery of digital banking services across the EDBI group.

Data Scientist

- Designed and optimized the Retrieval-Augmented Generation (RAG) pipeline for EximBot, an on-premises enterprise chatbot, using LangChain with Qdrant as the vector database, including document chunking, embedding generation, and retrieval workflows.
- Evaluated and integrated Vision-Language Models (Gemma Vision and Qwen2.5-VL) into the document ingestion pipeline to extract information from image-based PDF documents for enterprise knowledge retrieval.
- Developed an LLM-based agentic workflow for Farakhabar, automating news ingestion, translation, rewriting, competitor classification, storage in SQL and Qdrant, and banking impact analysis.
- Optimized prompt engineering strategies and retrieval workflows to improve response quality and reduce hallucinations .
- Fine-tuned Gemma 3 (27B) and Gemma 4 (31B) models using Hugging Face Transformers and parameter-efficient fine-tuning techniques (LoRA, QLoRA, PEFT) on NVIDIA A100 (80GB) GPUs with domain-specific financial datasets.
- Containerized AI services using Docker and deployed local LLM inference pipelines for secure on-premises enterprise environments.

ALZAHRA UNIVERSITY AI LAB, Tehran, Iran (Full-Time)

June 2024-April 2025

Alzahra University's AI Lab focuses on applying artificial intelligence and machine learning techniques to solve real-world problems in natural language processing and time-series forecasting.

AI Researcher

- Developed deep learning models for NLP and time-series forecasting, including demand forecasting for retail datasets using LSTM, GRU, and Transformer-based architectures.
- Conducted research on graph-based time-series forecasting using GPT-2, incorporating retrieval-augmented graph representations to improve short-term power forecasting accuracy.
- Developed an automatic speech-to-text pipeline using Whisper for a hospital information system, enabling physicians' voice recordings to be transcribed into structured clinical text for electronic medical records.

AFARINESH, Tehran, Iran (Full-Time)

January 2023-May 2024

Afarinеш is a knowledge-based IT firm, providing services to various startups in the form of a technical team as a service (TaaS), as well as developing several successful startups, such as Sayeh platform, Behkhaan social network, and book-selling startup Boxel.

Data Scientist

- Developed, optimized, and evaluated recommender systems for a book-focused social platform, including Neural Collaborative Filtering (NCF), Matrix Factorization (SVD), and Non-negative Matrix Factorization (NMF).
- Collaborated with backend engineers to integrate recommendation models into a C#/.NET API for production use.
- Performed large-scale data preprocessing, SQL-based data extraction, and A/B testing to evaluate recommender system performance.

TEACHING EXPERIENCE

Instructor: Electronics course and its general laboratory using Proteus Atrak Institute of Higher Education	September 2021-January 2022
Instructor: Microcontroller and microprocessor lab using CodeVisionAVR and Atmel Studio	September 2021-January 2022
Teaching Assistant: Electric circuits and logic circuit course Class of Dr. Ramin Rajaei	October 2018 -February 2019

SKILLS & LANGUAGES

Skills: Machine Learning Data Visualization Large Language Model (LLM) Data Structures and Algorithms Model Deployment SQL
Programming Languages: Python, C#, C VHDL Frameworks: Tensorflow, Pytorch, Hugging Face, LangChain ,Matplotlib, Keras, .Net, Git, Docker
Languages: Persian (Native) English (Professional Working Proficiency)

PROJECTS

Link Prediction Approach to Recommender Systems	April 2023
The "link prediction" project in recommender systems uses algorithms to predict connections between users or items, improving user experience through personalized recommendations. Various machine learning techniques like graph models and GCN-GNNs are employed for accurate predictions.	
Customer Sentiment Analysis	September 2022
I have experience implementing deep learning techniques such as Recurrent Neural Networks (RNNs) and Transformer-based models like BERT for customer sentiment analysis. This involved effectively extracting sentiment-related features from textual data to understand customer feedback better and improve decision-making processes and customer satisfaction strategies.	
Classification of text messages with Deep Learning	August 2022
In this project, I aimed to develop a system capable of categorizing text messages based on various topics and contents. To achieve this, I utilized advanced deep learning techniques such as recurrent neural networks (RNNs) and transformer-based neural networks like BERT.	
Persian OCR with VHDL	December 2017
This project uses VHDL, a language for modeling and simulating digital logic circuits, to implement OCR (Optical Character Recognition) components for real-time performance on FPGA platforms. This allows for converting Persian text images into editable, searchable digital text with speed and efficiency.	

PAPERS

Journal Paper

- Alinia, Masoumeh, and Seyed Mohammad Hossein Hasheminejad. "DiSA-CF: A distance-integrated self-attention model for collaborative filtering in web service recommendation." Expert Systems with Applications (2024): 125223.
* This journal paper presents a novel model for collaborative filtering in web service recommendations by integrating distance information and self-attention mechanisms to enhance the accuracy of quality of service predictions.

Conference Papers

- Alinia, Masoumeh. Hasheminejad, Seyed Mohammad Hossein. Shakibian, Hadi. "Link Prediction for Recommendation based on Complex Representation of Items Similarities." 13th International Conference on Computer and Knowledge Engineering. 2023.
* This research is a part of my PhD thesis, titled "Complex Dynamical Networks", later served as a foundation for further in-depth research on my thesis.
- Alinia, Masoumeh, Hasheminejad, Seyed Mohammad Hossein. "Location-Based Deep Collaborative Filtering for Quality-of-Service Prediction in IoT." 7th International Conference on Internet of Things and Applications (IoT). 2023.
* This conference paper presents a location-based deep collaborative filtering method for predicting service quality. By integrating location information into the collaborative filtering framework, the model improves the accuracy of service quality predictions.

OTHER RELATED COURSES TAKEN

ClassVision: Image processing and machine vision with OpenCV Graph Neural Network
Maktabkhooneh: Deep learning with TensorFlow and Keras Design and development of recommender systems Advanced TensorFlow and deep learning: Sequence Modeling Generative AI Training with Large Language Models (LLMs) Advanced Vision-Language Models (VLMs) Docker Training for Developers and DevOps Engineers Comprehensive MLOps Bootcamp with 10+ Hands-on Machine Learning Projects